# TR5 Worked Examples: Exact Trig Values

Recording companion for the TR5 CheckIt support video — Math 172, Santa Ana College

## How to use this document

- Each worked example is on its own page, in the same order as the Try It tabs in the interactive explainer, with the same numbers.
- The right-hand third of every page is intentionally blank. It is space for handwriting or ink notation while you narrate (for example, after inserting this PDF as a printout in OneNote).
- Say it at the start of the video: your CheckIt problem will have different numbers than the examples here. The reasoning is what transfers.

## The one question behind every example

**Where is the point on the unit circle, and what are its coordinates?** On the unit circle, $\cos\theta = x$ and $\sin\theta = y$, so $\tan\theta = y/x$, $\sec\theta = 1/x$, $\csc\theta = 1/y$ and $\cot\theta = x/y$. For a special angle, the reference angle gives the size of the coordinates and the quadrant gives their signs. From one coordinate, $x^2 + y^2 = 1$ gives the other, with its sign from the quadrant. A zero in a denominator means the value is undefined.

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# Example 1: Sine of 150 Degrees

$$\sin(150°)$$

## Question

What is the exact value?

- **A.** $-1/2$
- **B.** $(\sqrt{3})/2$
- **C.** $1/2$
- **D.** $-(\sqrt{3})/2$

## Answer and reasoning

150° is in Quadrant II, 30° short of 180°, so its reference angle is 30°. $\sin 30° = 1/2$, and sine (the *y*-coordinate) is positive in Quadrant II: $\sin 150° = 1/2$. $-1/2$ gives the wrong sign; $(\sqrt{3})/2$ is $\sin 60°$, from using the wrong reference angle or the cosine of 30°; and $-(\sqrt{3})/2$ makes both slips. ***1/2*.**

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# Example 2: Cosine of 5π/4

$$\cos(5\pi/4)$$

## Question

What is the exact value?

- **A.** $-(\sqrt{2})/2$
- **B.** $(\sqrt{2})/2$
- **C.** $-\sqrt{2}$
- **D.** $\sqrt{2}$

## Answer and reasoning

$5\pi/4=225°$, in Quadrant III, with reference angle $\pi/4$. $\cos\pi/4=(\sqrt{2})/2$, and cosine (the $x$-coordinate) is negative in Quadrant III: $-(\sqrt{2})/2$. $(\sqrt{2})/2$ misses the sign; $-\sqrt{2}$ is $1/x$, the secant; and $\sqrt{2}$ makes both slips. **$-(\sqrt{2})/2$.**

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# Example 3: Tangent of 2π/3

$$\tan(2\pi/3)$$

## Question

What is the exact value?

- **A.** $\sqrt{3}$
- **B.** $-\sqrt{3}$
- **C.** $-(\sqrt{3})/3$
- **D.** $(\sqrt{3})/3$

## Answer and reasoning

$2\pi/3 = 120°$, in Quadrant II, with reference angle 60°. The point is $(-1/2, (\sqrt{3})/2)$, so $\tan = y/x = (\sqrt{3}/2)/(-1/2) = -\sqrt{3}$. $\sqrt{3}$ misses that $x$ is negative; $-(\sqrt{3})/3$ is $x/y$, the cotangent, or $\tan 30°$ with a minus; and $(\sqrt{3})/3$ makes both slips. **$-\sqrt{3}$.**

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# Example 4: Secant of 270 Degrees

$$\sec(270°)$$

## Question

What is the exact value?

**A.** Undefined
**B.** 0, the x-coordinate
**C.** 1
**D.** −1, the y-coordinate

## Answer and reasoning

At 270° the point on the unit circle is $(0,-1)$, straight down. $\sec = 1/x = 1/0$, and dividing by 0 is undefined. CheckIt accepts "undefined" as an answer here. 0 is $\cos 270°$, the $x$-coordinate itself; 1 would be $\sec 0°$; and $-1$ is $\sin 270°$, or $\csc 270°$. **Undefined.**

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# Example 5: Tangent from a Point

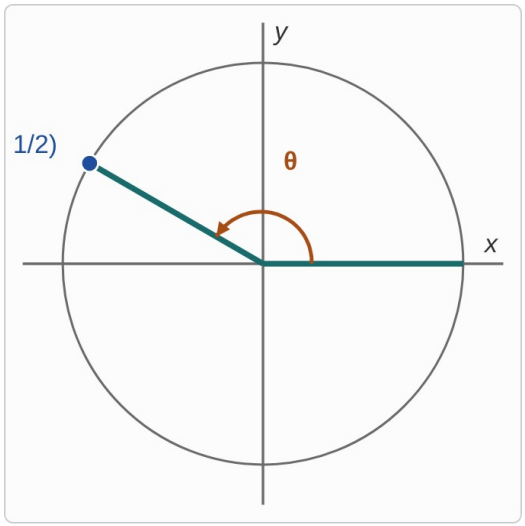



**Figure:** A unit circle with an angle θ in standard position. The terminal side points up and to the left, into Quadrant II, and meets the circle at the point labeled (−√3/2, 1/2).

$$(-(\sqrt{3})/2, 1/2)$$

## Question

The terminal side of θ meets the unit circle at this point. What is tan θ?

- **A.** $-\sqrt{3}$
- **B.** $-(\sqrt{3})/3$
- **C.** $(\sqrt{3})/3$
- **D.** $\sqrt{3}$

## Answer and reasoning

$\tan\theta = y/x = (1/2)/(-\sqrt{3}/2) = -1/\sqrt{3} = -(\sqrt{3})/3$. The halves cancel, and multiplying top and bottom by $\sqrt{3}$ clears the root from the bottom. $-\sqrt{3}$ is $x/y$, the cotangent; $(\sqrt{3})/3$ drops the sign of $x$; and $\sqrt{3}$ makes both slips. ***−(√3)/3*.**

# Example 6: Cosecant from a Point

$$((\sqrt{2})/2, -(\sqrt{2})/2)$$

## Question

The terminal side of θ meets the unit circle at this point. What is csc θ?

- **A.** $\sqrt{2}$
- **B.** $-(\sqrt{2})/2$
- **C.** $(\sqrt{2})/2$
- **D.** $-\sqrt{2}$

## Answer and reasoning

$\csc\theta = 1/y = 1/(-\sqrt{2}/2) = -2/\sqrt{2} = -\sqrt{2}$. $\sqrt{2}$ drops the sign of $y$; $-(\sqrt{2})/2$ is $y$ itself, the sine, without taking the reciprocal; and $(\sqrt{2})/2$ is $x$, the cosine. **$-\sqrt{2}$.**

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# Example 7: Secant from a Point

$$(-1/2, -(\sqrt{3})/2)$$

## Question

The terminal side of θ meets the unit circle at this point. What is sec θ?

- **A.** $-2$
- **B.** $2$
- **C.** $-(2\sqrt{3})/3$
- **D.** $-1/2$

## Answer and reasoning

$\sec\theta = 1/x = 1/(-1/2) = -2$. 2 drops the sign; $-(2\sqrt{3})/3$ is $1/y$, the cosecant; and $-1/2$ is $x$ itself, the cosine, without the reciprocal. **$-2$.**

# Example 8: On the Axis

$$(0,-1)$$

## Question

The terminal side of θ meets the unit circle at this point. What is cot θ?

**A.** −1
**B.** 0
**C.** Undefined
**D.** 1

## Answer and reasoning

cot$\theta = x/y = 0/(-1) = 0$. Zero on top is fine; only zero on the bottom is undefined. Undefined is tan$\theta = y/x = (-1)/0$, the reciprocal; $-1$ is $y$, the sine; and 1 has no source here except dropping the sign of $-1$. **0.**

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# Example 9: Quadrant II

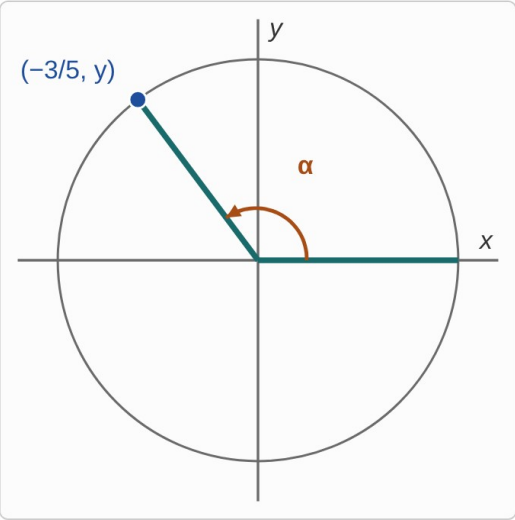



**Figure:** A unit circle with an angle α in standard position. The terminal side points up and to the left, into Quadrant II, and meets the circle at the point labeled (−3/5, y).

## Question

The point (x, y) on the unit circle is in Quadrant II with x = −3/5, and α is its angle. What is tan α?

- **A.** $-4/3$
- **B.** $4/3$
- **C.** $-3/4$
- **D.** $3/4$

## Answer and reasoning

$y^2=1-9/25=16/25$, so $y=\pm 4/5$, and in Quadrant II $y$ is positive: $y=4/5$. Then $tanalpha=y/x=(4/5)/(-3/5)=-4/3$. It is the 3, 4, 5 triple. $4/3$ drops the sign of $x$ (or makes $y$ negative too); $-3/4$ is $x/y$, the cotangent; and $3/4$ makes both slips. ***−4/3*.**

# Example 10: Quadrant IV

$$y=-5/13$$

## Question

The point (x, y) on the unit circle is in Quadrant IV with this y-value, and α is its angle. What is csc α?

**A.** *13/5*
**B.** *−5/13*
**C.** *−13/5*
**D.** *−13/12*

## Answer and reasoning

*cscalpha=1/y=1/(−5/13)=−13/5*. You do not need *x* for this one, though it is *12/13* (5, 12, 13, positive in Quadrant IV). *13/5* drops the sign; *−5/13* is *y* itself, the sine; and *−13/12* takes the reciprocal of *x*, the secant, and also gives it the wrong sign. ***−13/5*.**

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# Example 11: Quadrant III

$$x=-8/17$$

## Question

The point (x, y) on the unit circle is in Quadrant III with this x-value, and α is its angle. What is cot α?

- **A.** $-8/15$
- **B.** $15/8$
- **C.** $-15/8$
- **D.** $8/15$

## Answer and reasoning

$y^2=1-64/289=225/289$, so $y=-15/17$, negative in Quadrant III (the 8, 15, 17 triple). Then $cotalpha=x/y=(-8/17)/(-15/17)=8/15$: two negatives make a positive, as cotangent always is in Quadrant III. $-8/15$ makes $y$ positive; $15/8$ is $y/x$, the tangent; and $-15/8$ makes both slips. ***8/15*.**

# Example 12: Quadrant I

$$y=20/29$$

## Question

The point (x, y) on the unit circle is in Quadrant I with this y-value, and α is its angle. What is sec α?

**A.** *29/21*
**B.** *21/29*
**C.** *29/20*
**D.** *20/21*

## Answer and reasoning

$x^2=1-400/841=441/841$, so $x=21/29$, positive in Quadrant I (the 20, 21, 29 triple). Then $secalpha=1/x=29/21$. *21/29* is *x* itself, the cosine; *29/20* is *1/y*, the cosecant; and *20/21* is *y/x*, the tangent. ***29/21*.**

# Example 13: Slip 1

$$\cos(210°)=\cos(30°)=(\sqrt{3})/2$$

## Question

A student found this exact value. What is the mistake, if any?

**A.** There is no mistake.
**B.** In Quadrant III, x is negative: $-\sqrt{3}/2$.
**C.** The reference angle is 60°, so the value is 1/2.
**D.** Cosine of 210° is undefined, since 210° > 180°.

## Answer and reasoning

The reference angle is right: $210°-180°=30°$, so the size is $(\sqrt{3})/2$. But 210° is in Quadrant III, where the $x$-coordinate is negative, so $\cos 210°=-(\sqrt{3})/2$. The reference angle is measured to the $x$-axis, so it is 30°, not 60°; and cosine is defined for every angle. ***−(√3)/2*.**

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# Example 14: Slip 2

$$\tan(\pi/2)=\sin(\pi/2)/\cos(\pi/2)=1/0=0$$

## Question

A student found this exact value. What is the mistake, if any?

**A.** There is no mistake.
**B.** The cosine of π/2 is 1, so the value is 0/1 = 0.
**C.** Dividing by 0 is not 0: tan(π/2) is undefined.
**D.** 1/0 is 1, so tan(π/2) = 1.

## Answer and reasoning

At $\pi/2$ the point is $(0,1)$, so $\tan=y/x=1/0$, and division by 0 has no value: $\tan\pi/2$ is undefined. CheckIt's key prints undefined for this. $\cos\pi/2=0$, not 1, and $\sin\pi/2=1$; and $1/0$ is not 1 or anything else. **Undefined.**

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# Example 15: Slip 3

$$\csc(7\pi/4)=1/(-\sqrt{2}/2)=-\sqrt{2}$$

## Question

A student found this exact value. What is the mistake, if any?

- **A.** There is no mistake.
- **B.** Cosecant uses x: the value is √2.
- **C.** 7π/4 is in Quadrant III, so the value is √2.
- **D.** The answer must be rationalized: −√2/2.

## Answer and reasoning

$7\pi/4=315°$, in Quadrant IV, with point $((\sqrt{2})/2,-(\sqrt{2})/2)$. $\csc=1/y=-2/\sqrt{2}=-\sqrt{2}$. The student is right. Cosecant is the reciprocal of $y$, and $1/x$ is secant; $7\pi/4$ is between $3\pi/2$ and $2\pi$, Quadrant IV; and $-\sqrt{2}$ has no root in a denominator, while $-(\sqrt{2})/2$ is the sine, a different value. **There is no mistake.**

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# Example 16: Slip 4

$$(-1/2,(\sqrt{3})/2): \quad \sec\theta = 2/\sqrt{3} = (2\sqrt{3})/3$$

## Question

A student found sec θ for the angle whose unit-circle point is shown. What is the mistake, if any?

**A.** There is no mistake.
**B.** It should not be rationalized; leave it as $2/\sqrt{3}$.
**C.** Secant is y/x: the value is $-\sqrt{3}$.
**D.** Secant is 1/x, so the value is $-2$.

## Answer and reasoning

$\sec\theta = 1/(\cos\theta) = 1/x = 1/(-1/2) = -2$. The student used $1/y$, which is the cosecant. Rationalizing is fine and equal, but it is the wrong function either way; and $y/x = -\sqrt{3}$ is the tangent. **$-2$.**

# Example 17: Slip 5

$$y=12/13,\ \text{Quadrant II}$$
$$x=\sqrt{(1-144/169)}=5/13$$
$$\cos\alpha=5/13$$

## Question

A student found cos α for a unit-circle point in Quadrant II. What is the mistake, if any?

**A.** There is no mistake.
**B.** In Quadrant II, x is negative: −5/13.
**C.** cos α is the y-value, so it is 12/13.
**D.** x = 1 − 12/13 = 1/13, so cos α = 1/13.

## Answer and reasoning

The size of $x$ is right: $\sqrt{(25/169)}=5/13$. But the square root gives only the size, and in Quadrant II $x$ is negative, so $x=-5/13$ and $\cos\alpha=-5/13$. Cosine is the $x$-value, and $12/13$ is the sine; and $x^2+y^2=1$ works with squares, so $1-12/13$ is not $x$. ***−5/13*.**